

UJJWAL KUMAR

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VLSI and Embedded Systems undergraduate with hands-on RTL-to-GDSII ASIC tape-out experience (Cadence Genus/Innovus, 90nm CMOS) and embedded firmware development across ESP32, AVR, and Raspberry Pi platforms. Comfortable across the stack from gate-level timing closure to bare-metal C.

EDUCATION

Indian Institute of Information Technology, Manipur — B.Tech, Electronics & Communication Engineering (VLSI & Embedded Systems)

Aug 2023 – Dec 2027 | CGPA: 8.02

TECHNICAL SKILLS

HDL & RTL: Verilog HDL, SystemVerilog, HLS (C/C++), Pipeline Design, FSM, Hazard Detection

ASIC & FPGA: Cadence Genus, Cadence Innovus, Xilinx Vivado, RTL-to-GDSII, STA, DRC/LVS, LTspice

Embedded & Firmware: ESP32, ATmega2560, Raspberry Pi, Pixhawk, ArduPilot, Bare-Metal AVR, ISR Design

Protocols & IoT: UART, SPI, I2C, BLE, CAN Bus, MQTT, Wi-Fi

Programming: C, C++, Embedded C, Python, TCL, Shell, Perl

TECHNICAL PROJECTS

CSHM FIR Filter — RTL-to-GDSII ASIC [GitHub](#)

Verilog · Cadence Genus · Cadence Innovus · 90nm CMOS

- Took an 11-tap FIR filter through a complete RTL-to-GDSII flow on 90nm CMOS — floorplanning, placement, CTS, routing, and DRC/LVS sign-off. Reached 200 MHz with an 80% power reduction and a fully DRC-clean layout.

5-Stage Pipelined RV32I RISC-V Processor [GitHub](#)

Verilog · Xilinx Vivado · Zybo Z7 · TCL

- Implemented the full RV32I ISA with dual forwarding paths, load-use hazard detection, and branch resolution in EX with a 2-instruction flush. WNS +5.174ns @ 125MHz, self-checking testbench, 0 errors.

HLS-Based 8-Operation ALU

Vitis HLS · C++ · Xilinx Artix-7 · Vivado

- Designed an ALU in C++ via Vitis HLS, pipelined for II=1 throughput, implemented on an Artix-7 FPGA with zero timing violations.

ESP32 Smart Home Automation [GitHub](#)

ESP32 · MQTT · BLE · Arduino IoT Cloud · HiveMQ

- Built a 4-relay home automation system with three simultaneous control modes (cloud, BLE/IR remote, manual switches) kept in real-time bidirectional sync via dual MQTT stacks.

Bare-Metal Game Engine — ATmega2560 [GitHub](#)

Embedded C · AVR ISR · CGRAM

- Wrote a 16×2 LCD game in bare-metal Embedded C with direct register manipulation (no HAL), a Timer1 ISR game loop, and custom animated sprites via CGRAM encoding.

Raspberry Pi Surveillance System [GitHub](#)

Raspberry Pi 3 · Python · libcamera · Telegram Bot API

- Built an end-to-end IoT surveillance pipeline with real-time cross-network image transmission and an auto-restart watchdog for unattended uptime.

CERTIFICATIONS

NIELIT Virtual Academy — Online Internship, RTL Design using Verilog HDL (Cert. NVA-C-IN13-12-5218), May–Jul 2026

NIELIT Virtual Academy — Online Internship, HLS Programming, C/C++ based HDL Design (Cert. NVA-C-IN14-12-5243), May–Jul 2026

IIT (ISM) Dhanbad — Bootcamp, Electronics for Unmanned Aerial Systems, Jul 2025

ACHIEVEMENTS

2nd Place, Circuit Design Competition — NIELIT Imphal, 2025

3rd Place, Line Follower Robot & Innovation Expo — Manipur Technical University, 2025